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**PHYSICS**

**9702/22**

Paper 2 AS Level Structured Questions

**May/June 2019**

MARK SCHEME

Maximum Mark: 60

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **10** printed pages.



**Cambridge Assessment**  
International Education

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

| Question | Answer                                                                                                                              | Marks     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1(a)     | absolute uncertainty = $(1.6 / 100) \times 0.0125$<br>$= 2 \times 10^{-4} \text{ m}$                                                | <b>A1</b> |
| 1(b)(i)  | $p = (4 \times 0.38) / (\pi \times 0.0125^2)$<br>$= 3100 \text{ N m}^{-2}$                                                          | <b>A1</b> |
| 1(b)(ii) | percentage uncertainty = $2.8 + (2 \times 1.6)$ (= 6%)<br><b>or</b><br>fractional uncertainty = $0.028 + (2 \times 0.016)$ (= 0.06) | <b>C1</b> |
|          | absolute uncertainty = $0.06 \times 3100$<br>$= 190 \text{ N m}^{-2}$ ( <i>allow to 1 significant figure</i> )                      | <b>A1</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 2(a)     | (resultant) force proportional/equal to/is rate of change of momentum                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>B1</b> |
| 2(b)(i)  | <p>distance = area under graph <b>or</b> <math>s = \frac{1}{2}(u + v)t</math><br/> <math>= \frac{1}{2} \times (9 + 13) \times 10</math></p> <p><b>or</b></p> <p><math>s = ut + \frac{1}{2}at^2</math><br/> <math>= (9 \times 10) + (\frac{1}{2} \times 0.40 \times 10^2)</math></p> <p><b>or</b></p> <p><math>s = vt - \frac{1}{2}at^2</math><br/> <math>= (13 \times 10) - (\frac{1}{2} \times 0.40 \times 10^2)</math></p> <p><b>or</b></p> <p><math>v^2 = u^2 + 2as</math><br/> <math>13^2 = 9^2 + (2 \times 0.40 \times s)</math></p> | <b>C1</b> |
|          | distance = 110 m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>A1</b> |

| Question | Answer                                                                                                                                            | Marks     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 2(b)(ii) | 1. $a = \text{gradient}$ <b>or</b> $a = (v - u) / t$ <b>or</b> $a = \Delta v / (\Delta)t$<br>e.g. $a = (14 - 9) / 12.5$ <b>or</b> $(13 - 9) / 10$ | <b>C1</b> |
|          | $a = 0.40 \text{ m s}^{-2}$                                                                                                                       | <b>A1</b> |
|          | 2. resultant force $= 850 \times 0.40$<br>$= 340 \text{ N}$                                                                                       | <b>A1</b> |
|          | 3. $(F =) 510 + 440 + 340 = 1300 \text{ (N)}$                                                                                                     | <b>A1</b> |
|          | 4. $P = Fv$                                                                                                                                       | <b>C1</b> |
|          | $= 1300 \times 13$<br>$= 1.7 \times 10^4 \text{ W}$                                                                                               | <b>A1</b> |
| 2(c)     | $E = \sigma / \epsilon$                                                                                                                           | <b>C1</b> |
|          | $E = (F / A) / (\Delta L / L)$ <b>or</b> $E = FL / A\Delta L$                                                                                     | <b>C1</b> |
|          | $\Delta L = (480 \times 0.48) / (3.0 \times 10^{-4} \times 2.2 \times 10^{11})$<br>$= 3.5 \times 10^{-6} \text{ m}$                               | <b>A1</b> |
| 2(d)     | $f_o = f_s v / (v - v_s)$<br>$480 = f_s \times 340 / (340 - 14)$                                                                                  | <b>C1</b> |
|          | $f_s = 460 \text{ Hz}$                                                                                                                            | <b>A1</b> |

| Question  | Answer                                                                                                                                                                      | Marks     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(a)      | the point where (all) the weight (of the body) is taken to act                                                                                                              | <b>B1</b> |
| 3(b)(i)   | vertical component = $54 \sin 35^\circ$<br>= 31 N                                                                                                                           | <b>A1</b> |
| 3(b)(ii)  | the (line of action of the) force (at B) passes through (point) A<br><b>or</b><br>the (line of action of the) force (at B) has zero (perpendicular) distance from (point) A | <b>B1</b> |
| 3(b)(iii) | $54 \sin 35^\circ \times 0.68$ <b>or</b> $54 \cos 35^\circ \times 0.68$ <b>or</b> $W \times 0.34$                                                                           | <b>C1</b> |
|           | $54 \sin 35^\circ \times 0.68 + 54 \cos 35^\circ \times 0.68 = W \times 0.34$ so $W = 150$ (N)                                                                              | <b>A1</b> |
| 3(b)(iv)  | total vertical force = $150 - 31$<br>= 120 N                                                                                                                                | <b>A1</b> |
| 3(c)      | $(\Delta)E = mg(\Delta)h$                                                                                                                                                   | <b>C1</b> |
|           | $E = \frac{1}{2}mv^2$                                                                                                                                                       | <b>C1</b> |
|           | ratio = $(m \times 9.81 \times 4.8) / (\frac{1}{2} \times m \times 9.2^2)$ <b>or</b> $(9.81 \times 4.8) / (\frac{1}{2} \times 9.2^2)$                                       | <b>C1</b> |
|           | = 1.1                                                                                                                                                                       | <b>A1</b> |

| Question  | Answer                                                                                                                                                      | Marks     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)(i)   | distance (in a specified direction of particle/point on wave) from the equilibrium position                                                                 | <b>B1</b> |
| 4(a)(ii)  | the maximum distance (of particle/point on wave) from the equilibrium position<br><br><b>or</b><br><br>the maximum displacement (of particle/point on wave) | <b>B1</b> |
| 4(b)      | $I \propto A^2$                                                                                                                                             | <b>C1</b> |
|           | $I_R / I = (3.6 - 1.2)^2 / (1.2)^2$<br><br>resultant intensity = $4.0I$                                                                                     | <b>A1</b> |
| 4(c)(i)   | as wave(s) pass through the slit(s)                                                                                                                         | <b>B1</b> |
|           | wave(s) spread (into geometric shadow)                                                                                                                      | <b>B1</b> |
| 4(c)(ii)  | $n\lambda = d \sin \theta$                                                                                                                                  | <b>C1</b> |
|           | $3\lambda = d \sin 90^\circ$ <b>or</b> $3\lambda = d$                                                                                                       | <b>C1</b> |
|           | $d = 3 \times 630 \times 10^{-9}$<br><br>$= 1.9 \times 10^{-6} \text{ m}$                                                                                   | <b>A1</b> |
| 4(c)(iii) | wavelength of blue light is shorter (than 540 nm/630 nm/wavelengths of original light)                                                                      | <b>M1</b> |
|           | (so) third order diffraction maximum is produced                                                                                                            | <b>A1</b> |

| Question  | Answer                                                                                                           | Marks     |
|-----------|------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)      | <u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s)                                                                  | <b>M1</b> |
|           | around a loop/around a closed circuit                                                                            | <b>A1</b> |
| 5(b)(i)   | 1. $1/R = 1/R_1 + 1/R_2$<br>$1/R = 1/90 + 1/18$                                                                  | <b>C1</b> |
|           | $R = 15\ \Omega$                                                                                                 | <b>A1</b> |
|           | 2. $I = V/R$                                                                                                     | <b>C1</b> |
|           | $I = 4.8/15$ <b>or</b> $I = 4.8/90 + 4.8/18$<br>$I = 0.32\text{ A}$                                              | <b>A1</b> |
| 5(b)(ii)  | $E = V + Ir$<br><b>or</b><br>$E = I(R + r)$                                                                      | <b>C1</b> |
|           | $5.6 = 4.8 + 0.32\ r$ so $r = 2.5\ (\Omega)$<br><b>or</b><br>$5.6 = 0.32 \times (15 + r)$ so $r = 2.5\ (\Omega)$ | <b>A1</b> |
| 5(b)(iii) | $P = EI$ <b>or</b> $P = VI$ <b>or</b> $P = I^2R$ <b>or</b> $P = V^2/R$                                           | <b>C1</b> |
|           | ratio = $(0.32^2 \times 2.5) / (5.6 \times 0.32)$ <b>or</b> $0.256 / 1.792$                                      | <b>C1</b> |
|           | = 0.14                                                                                                           | <b>A1</b> |

| Question | Answer                        | Marks     |
|----------|-------------------------------|-----------|
| 5(c)     | $7.2 - 5.6 - 2.5I - 3.5I = 0$ | <b>C1</b> |
|          | $I = 0.27 \text{ A}$          | <b>A1</b> |

| Question | Answer                                                                                      | Marks     |
|----------|---------------------------------------------------------------------------------------------|-----------|
| 6(a)     | path/direction in which a (free) <u>positive</u> charge will move                           | <b>B1</b> |
| 6(b)     | (lines) closer together in Y/further apart in X                                             | <b>B1</b> |
| 6(c)(i)  | $a = Eq / m$<br><br><b>or</b><br>$F = Eq$ <b>and</b> $F = ma$                               | <b>C1</b> |
|          | ratio = $(1e / 0.15 \text{ u}) \times (4 \text{ u} / 2e)$ <b>or</b> $1 / 0.15 \times 4 / 2$ | <b>C1</b> |
|          | ratio = 13                                                                                  | <b>A1</b> |
| 6(c)(ii) | down quark charge is $-(1/3)e$                                                              | <b>C1</b> |
|          | $-(1/3)e + q = -1e$ so $q = -(2/3)e$                                                        | <b>A1</b> |
|          | $-(2/3)e$ is anti-up / $\bar{u}$ (quark) (allow charm or top antiquark)                     | <b>B1</b> |